# OLEKSANDR SHTYKHNO

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## SUMMARY

Software Engineer — 25+ years across backend, event-driven data platforms and cloud-native architecture (Java, Scala, Kotlin, Golang, Kubernetes, AWS/Azure/GCP), currently engineering AI-augmented delivery at ING inside a policy-locked enterprise environment.

## KEY IMPACT

- **Cloud migration:** 35+ microservices (60+ repos), 5+ databases, Kafka, Cassandra and Elasticsearch — AWS → Azure with **zero downtime**.
- **Scale:** track-and-trace read path rebuilt on Cassandra + Elasticsearch — **3–4 s → 15–30 ms at ~1,000 rps**; the top customer complaint resolved.
- **Modernisation & delivery:** Scala 2.13 → 3 estate-wide; 30+ endpoints Akka HTTP → ZIO; a Scala service rewritten in Go on a **2-month deadline**.
- **AI-augmented engineering:** agent harnesses, session-hook memory and RAG knowledge bases — engineered around a policy-locked environment (no MCP, no vendor memory).

## CORE STACK

- **Languages & FP:** Scala 2/3 (8+ yrs — Cats, ZIO, Akka), Java (15+ yrs), Go, Kotlin; functional design, TDD.
- **Data & Streaming:** Kafka (upgrade, schema registry), Cassandra, PostgreSQL, Oracle, Elasticsearch, Hadoop, DynamoDB, S3 / Azure Blob.
- **Cloud & Platform:** AWS, Azure, Kubernetes, Docker, GitHub Actions, Jenkins, gRPC / REST, DataDog.
- **AI & Agentic:** GitHub Copilot CLI (GH-300), Skills and session hooks, agent harnesses, RAG / LLM knowledge bases, prompt engineering, MCP.
- **Architecture:** microservices, event-driven design, zero-downtime migration, enterprise architecture, Agile / CI-CD, mentoring.

## RECENT WORK EXPERIENCE

### Software Engineer, ING, Netherlands | Nov 2025 - Present

External consultant — Data Customer NL domain.

**Technology stack:** Scala 2, Cats, Kafka, Postgres, Oracle, GitHub Copilot CLI.

- **AI coach and presenter for the team:** ran hands-on sessions on Copilot CLI, custom Skills and agent workflows, and coached engineers to apply AI-assisted development themselves.
- Gave a locked-down coding assistant persistent cross-session memory with a second-brain layer of session hooks, Skills and Markdown storage — no MCP, near-zero maintenance — and turned scattered team documentation into a searchable RAG wiki.
- Cut integration overhead with a shared agent-as-runtime library reused across team projects, automated PACT integration testing via Simulation TestKit, and cut technical debt in a business-critical Scala 2 application by removing an outdated stabilizer module and supporting its decommissioning.

### Freelance Scala/Go Engineer, Shell Recharge, Netherlands | Nov 2024 - Nov 2025

Assignment in the MSP (Mobility Service Provider) domain.

**Technology stack:** Scala 2/3, GoLang, Cats, ZIO, Akka, Kafka, Postgres, DynamoDB, AWS, K8s, DataDog.

- Restored message throughput and reliability by diagnosing and resolving critical Kafka consumer bottlenecks throttling production processing.
- Modernised services across stacks: 30+ endpoints moved from Akka HTTP to ZIO behind a test safety net, and a Scala microservice rewritten in Go (30+ endpoints) and shipped to production **within a 2-month deadline**, cutting its resource footprint for handover to the offshore team.
- Built a universal rate limiter as ZIO middleware, fixed Kubernetes rollback procedures causing failed deployments, and de-risked the charging-card migration to a third-party system.

### Freelance Scala Engineer / Solution Architect, FedEx Express, Netherlands | Jun 2018 - Nov 2024

Commerce-critical B2B platform; stakeholder coordination, coaching and technical reviews.

**Technology stack:** Scala, Cats, ZIO, Akka, Kafka, Cassandra, Postgres, Oracle, ElasticSearch, AWS, Azure, K8s.

- **Cloud migration, zero downtime:** Migrated the whole platform from AWS to Azure — **35+ microservices across 60+ repositories**, including 5+ PostgreSQL databases, a Kafka cluster, Cassandra and Elasticsearch — **with no customer-visible downtime**, in small verifiable steps behind feature flags; storage moved with no code change.
- **Kafka platform:** Upgraded to Kafka 2.x and moved the cluster across clouds with extra consumers and gradual producer switch-over — also without downtime.
- **Scale — track & trace (~1,000 rps):** Replaced a Node.js/PostgreSQL read path with Cassandra plus Elasticsearch, taking customer-facing queries from **3–4 seconds to 15–30 ms** at **~1,000 rps** and resolving the top customer complaint.
- **Latency:** Cut service-to-service latency from **30 ms to 5 ms** with a proto-definition library that moved internal calls from REST to gRPC.
- **Modernisation, CI & team:** Scala 2.13 → 3; Jenkins → GitHub Actions with custom pipelines and build images; Kubernetes migration surfacing JDK CPU/memory pitfalls; coached engineers, reviewed designs, ran technical interviews.

### Senior Software Engineer, Bol.com, Netherlands | Nov 2015 - Jun 2018

One of the Netherlands' largest e-commerce platforms — Java, Kotlin, Spring Boot, Hadoop, SQL/NoSQL, full-text search, GCP.

- Extended the platform's message-based and reactive architecture for higher product-data volumes, and delivered SQL, NoSQL and full-text search solutions.

### Senior Software Engineer, AssetControl, Netherlands | Sep 2010 - Nov 2015

Reference-data and risk-data software for financial institutions — Java, Spring, Oracle, JIRA plugin development, test automation.

- Delivered a risk-data manager for curve-based financial data, a real-time operations dashboard (desktop and mobile), a JIRA plugin adopted internally for Agile practice, and improved test automation for large data volumes.

## EDUCATION & CERTIFICATIONS

**Master's degree, Computer Engineering** — National Technical University of Ukraine 'Kharkiv Polytechnic Institute', 1997–2003

GH-300 GitHub Copilot (2026) · Prompt Engineering for AI Bootcamp (2025) · Software Architecture: Developer to Architect (2024) · Rust by Building Real Applications (2024) · Certified Kubernetes Application Developer (2020) · Scala Advanced (2018)

**Languages:** Ukrainian (native) · English (full professional) · Dutch (intermediate, B1)